Limits of sequences

Lecture and worked examples

Objectives

Distinguish convergence from divergence; compute a limit; use comparison, the squeeze theorem and monotone convergence.

$u_n \to \ell \in \mathbb{R}$ means that every open interval containing ℓ contains all the terms from some index onwards. We say that (u_n) converges to ℓ. $u_n \to +\infty$ means that for every real number A, there is an index N such that $n \geq N \Rightarrow u_n \geq A$. $u_n \to -\infty$ means that for every real number A, there is N such that $n \geq N \Rightarrow u_n \leq A$.

Course notes

A sequence may have no limit at all, not even an infinite one: $(-1)^n$ oscillates between -1 and 1. Every convergent sequence is bounded, but the converse is false.

Operations and comparisons

For two convergent sequences with limits a, b, the sum tends to $a + b$, the product to ab and the quotient to $\frac{a}{b}$ if $b \neq 0$. The rules involving infinity are used by keeping track of the signs. The forms $+\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$ do not allow a direct conclusion.

Operations and comparisons

$\frac{3n^2 - n + 1}{2n^2 + 4} = \frac{3 - \frac{1}{n} + \frac{1}{n^2}}{2 + \frac{4}{n^2}} \to \frac{3}{2}$. For $\sqrt{n^2 + n} - n$, multiplying by the conjugate expression gives $\frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \to \frac{1}{2}$.

Operations and comparisons

If $u_n \leq v_n$ from some index onwards and $u_n \to +\infty$, then $v_n \to +\infty$. If $a_n \leq u_n \leq b_n$ from some index onwards and $a_n, b_n \to \ell$, then $u_n \to \ell$ (squeeze theorem). Every increasing sequence that is bounded above converges. Every decreasing sequence that is bounded below converges. This theorem is admitted (taken without proof).

Operations and comparisons

For every A, the terms of u exceed A beyond an index N_1. The inequality $v_n \geq u_n$ holds beyond an index N_2. For $n \geq \max(N_1, N_2)$, $v_n \geq A$.

For every A, the fact that the sequence is not bounded above provides N such that $u_N > A$. Since the sequence is increasing, $n \geq N \Rightarrow u_n \geq u_N > A$, hence $u_n \to +\infty$. Not being bounded above is not enough on its own.

Geometric sequences

$$q^n \longrightarrow \begin{cases} 0 & \text{si } -1 < q < 1, \\ 1 & \text{si } q = 1, \\ +\infty & \text{si } q > 1. \end{cases}$$

If $q \leq -1$, (q^n) has no limit: alternation for $q = -1$; for $q < -1$, the even and odd subsequences have opposite signs and unbounded moduli.

Geometric sequences

Fix $a \geq 0$. Let us prove $P(n)$: "$(1+a)^n \geq 1+na$". At index 0, both sides equal 1. Fix $n \geq 0$ and assume $P(n)$ is true. Since $1+a > 0$,

$$(1+a)^{n+1} \geq (1+na)(1+a) = 1 + (n+1)a + na^2 \geq 1 + (n+1)a.$$

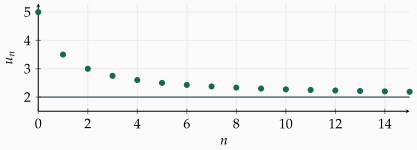
This is $P(n+1)$ because $na^2 \geq 0$. By induction, the inequality holds for every $n \geq 0$. If $q > 1$, take $a = q - 1 > 0$ and compare with $1 + na$. If $0 < q < 1$, apply this result to $\dfrac{1}{q}$, then take reciprocals. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ and the squeeze theorem concludes. The case $q = 0$ is handled from index 1 onwards.

Geometric sequences

For $u_{n+1} = f(u_n)$, establish a bound and monotonicity to prove convergence. If f is continuous at the limit ℓ, passing to the limit gives $\ell = f(\ell)$ (continuity is studied in chapter 4). Solving this equation never proves, on its own, that the sequence converges.

Geometric sequences

The points get closer to 2 without reaching it:
$u_n = 2 + \dfrac{3}{n+1}$.



Research activity

A limit is not a last term - 30 min.

Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = 2 + \dfrac{3}{n+1}$.

Research activity (2)

1. Compute u_0, u_9 and u_{99}. Is there an integer n such that $u_n = 2$? Justify.
2. Find the smallest integer N such that, for every integer $n \geq N$, $|u_n - 2| < 0.01$.
3. Let $\varepsilon > 0$. Give, in terms of ε, an integer N such that $|u_n - 2| < \varepsilon$ for every $n \geq N$.
4. Define on $\mathbb{N}$ the sequence (v_n) by $v_n = 2 + \dfrac{(-1)^n}{n+1}$. Prove that it converges to 2 and that it is not monotonic. What does this example show?